

```
library(ISLR)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(glmnet)
```

```
## Loading required package: Matrix

## Loaded glmnet 4.1-10
```

```
library(caret)
```

```
## Loading required package: ggplot2

## Warning: package 'ggplot2' was built under R version 4.5.2

## Loading required package: lattice
```

```
Carseats_Filtered <- Carseats %>%
  select(Sales, Price, Advertising, Population, Age, Income, Education)
```

```
library(ISLR)
library(dplyr)
library(caret)
```

```
set.seed(123)
```

```
Carseats_Filtered <- Carseats %>%
  select(Sales, Price, Advertising, Population, Age, Income, Education)
```

```
train_control <- trainControl(method = "cv", number = 5)
```

```
svm_linear_model <- train(
  Sales ~ .,
  data = Carseats_Filtered,
  method = "svmLinear",
  trControl = train_control
)
```

```
svm_linear_model
```

```
## Support Vector Machines with Linear Kernel
##
## 400 samples
## 6 predictor
##
## No pre-processing
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 320, 321, 319, 320, 320
## Resampling results:
##
## RMSE      Rsquared   MAE
## 2.305688  0.3442665  1.85399
##
## Tuning parameter 'C' was held constant at a value of 1
```

```
set.seed(123)

train_control_repeated <- trainControl(
  method = "repeatedcv",
  number = 5,
  repeats = 2
)

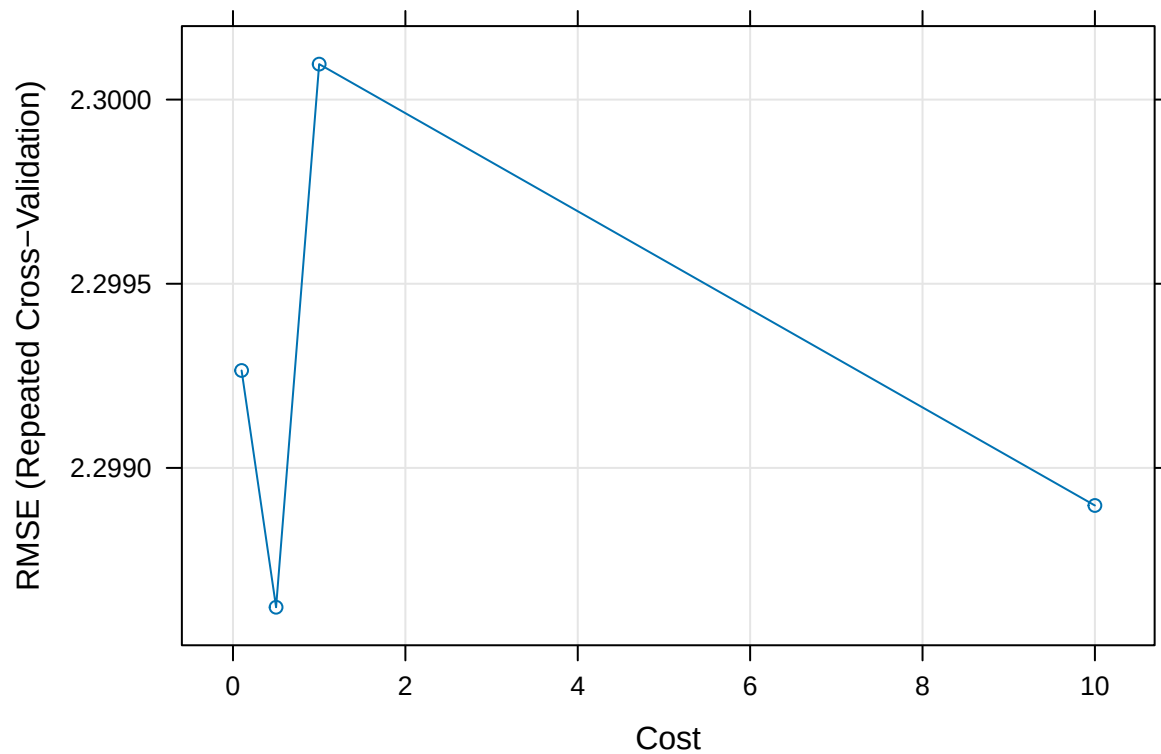
svm_grid <- expand.grid(C = c(0.1, 0.5, 1, 10))

svm_tuned_model <- train(
  Sales ~ .,
  data = Carseats_Filtered,
  method = "svmLinear",
  trControl = train_control_repeated,
  tuneGrid = svm_grid
)

svm_tuned_model
```

```
## Support Vector Machines with Linear Kernel
##
## 400 samples
## 6 predictor
##
## No pre-processing
## Resampling: Cross-Validated (5 fold, repeated 2 times)
## Summary of sample sizes: 320, 321, 319, 320, 320, 319, ...
## Resampling results across tuning parameters:
##
## C      RMSE      Rsquared   MAE
## 0.1    2.299265  0.3462103  1.840099
## 0.5    2.298622  0.3466721  1.840422
## 1.0    2.300096  0.3457092  1.841191
## 10.0   2.298898  0.3464189  1.840055
##
## RMSE was used to select the optimal model using the smallest value.
## The final value used for the model was C = 0.5.
```

```
plot(svm_tuned_model)
```



```
set.seed(123)
```

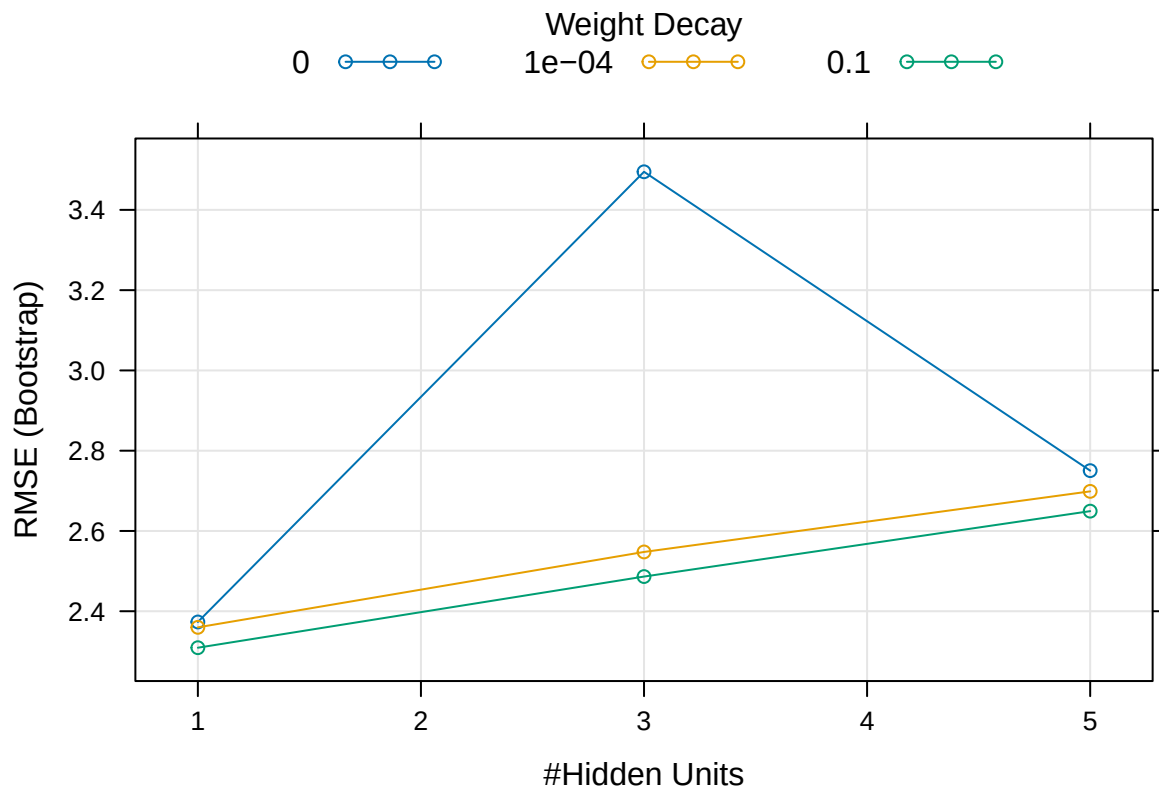
```
nnet_model <- train(  
  Sales ~ .,  
  data = Carseats_Filtered,  
  method = "nnet",  
  preProcess = c("center", "scale"),  
  linout = TRUE,  
  trace = FALSE,  
  maxit = 500  
)
```

```
nnet_model
```

```
## Neural Network  
##  
## 400 samples  
## 6 predictor  
##  
## Pre-processing: centered (6), scaled (6)  
## Resampling: Bootstrapped (25 reps)  
## Summary of sample sizes: 400, 400, 400, 400, 400, 400, ...  
## Resampling results across tuning parameters:
```

```
##
## size decay RMSE Rsquared MAE
## 1 0e+00 2.373259 0.3276650 1.879483
## 1 1e-04 2.359881 0.3236997 1.875530
## 1 1e-01 2.309112 0.3448199 1.844761
## 3 0e+00 3.494907 0.2224327 2.161130
## 3 1e-04 2.547740 0.2481713 2.039434
## 3 1e-01 2.486399 0.2694550 1.997456
## 5 0e+00 2.750304 0.1978340 2.204929
## 5 1e-04 2.698664 0.2119883 2.161521
## 5 1e-01 2.649375 0.2154909 2.140747
##
## RMSE was used to select the optimal model using the smallest value.
## The final values used for the model were size = 1 and decay = 0.1.
```

```
plot(nnet_model)
```



```
new_data <- data.frame(
  Price = 6.54,
  Advertising = 0,
  Population = 124,
  Age = 76,
  Income = 110,
  Education = 10
)
```

```
predicted_sales <- predict(nnet_model, newdata = new_data)
predicted_sales
```

```
##          1
## 11.46014
```